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DNN-Approximated Offset-Free MPC on FPGA for 10-kHz Constrained In-Plane Control of 6-DOF Magnetic Levitation System

Chang Hwan Kim, Jung Jun Kim, Hyeong Min Yoon, and Jun Young Yoon

***Abstract*—This article presents a deep neural network (DNN)-approximated offset-free MPC framework, implemented on an FPGA, for 10-kHz constrained in-plane control of a six-degree-of-freedom (6-DOF) magnetic levitation system. Classical motion controllers have been widely used in 6-DOF magnetic levitation systems, but their limited ability to explicitly handle input constraints may degrade tracking performance and compromise stability. Furthermore, state-constrained control is essential to prevent collisions and enforce velocity limits. Although MPC has recently been introduced for 6-DOF magnetic levitation systems, achieving a 10-kHz-level control rate required for high-precision motion control remains challenging due to its computational burden. To address this limitation, an MPC framework based on an augmented dynamics model is developed and tailored for the system to achieve offset-free tracking, input- and state-constrained control, and enhanced tracking performance. Subsequently, the control law is approximated by a DNN via supervised learning and deployed on an FPGA, shifting the computational burden of MPC optimization from online to offline computation. Experimental results demonstrate that the proposed framework significantly reduces the online computational cost and enables 10 kHz control with deterministic timing under input and state constraints. Comparative experiments further demonstrate offset-free control and improved tracking performance over conventional controllers and prior MPC implementations.**

***Index Terms*—Constrained control, deep neural network (DNN), FPGA deployment, high-precision motion control, magnetic levitation, model predictive control (MPC).**

I. Introduction

SIX-DEGREE-OF-FREEDOM (6-DOF) magnetic levitation systems have been widely used as positioning platforms in various high-precision manufacturing applications that demand contactless 6-DOF motion, such as semiconductor wafer stages [1], [2], microscope stages [3], and vibration isolation systems [4], [5]. By eliminating mechanical contact that can deteriorate positioning accuracy, these systems enable ultraprecise motion control [6]. In these systems, precise motion control relies not only on the mechatronic hardware but also critically on the control law.

Various classical motion control approaches, including PD controllers [4], PID controllers [1], [7], [8], [9], lead–lag compensators [2], [3], [6], and linear quadratic regulators (LQR) [10], [11], have been extensively utilized to achieve precise motion control of 6-DOF magnetic levitation systems. These linear controllers are straightforward to implement and can achieve a reasonable level of dynamic performance. Robust control approaches, such as sliding mode controllers (SMC), have been used to deal with model uncertainties and disturbances [12], [13]. Other control methods, including iterative learning control [14], [15] and fuzzy control [16], [17], have also been used to decouple DOFs and handle unmodeled dynamics. Nevertheless, these controllers do not explicitly handle input and state constraints, which may compromise system stability or degrade transient performance.

Model predictive control (MPC) is an attractive control strategy in this regard, as it explicitly handles input and state constraints while predicting future states to compute optimal control inputs. Owing to these advantages, MPC has been applied to a wide range of applications, including industrial processes [18], autonomous vehicles [19], quadrotors [20], robotics [21], and humanoid robots [22], ranging from slow to fast dynamics. Recently, MPC has also been employed for the motion control of 6-DOF magnetic levitation systems [23], [24], [25]. These systems typically operate under well-defined and relatively stable conditions, which may reduce model uncertainties and make them attractive for MPC-based control.

However, despite these advantages, applying MPC to magnetic levitation systems remains computationally challenging, as the optimization problem within the MPC framework must be solved in real time at each control cycle. This challenge is particularly critical for systems with fast and inherently unstable dynamics, which require control frequencies in the kilohertz to tens-of-kilohertz range, as insufficient sampling rates introduce phase lag that can destabilize the closed-loop system. Consequently, prior real-time MPC studies for 6-DOF magnetic levitation systems [23], [24], [25] have typically been implemented at 1 kHz and have mainly focused on input constraints, while state constraints have not been explicitly considered. Therefore,

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designing an MPC for 6-DOF magnetic levitation systems leads to the following requirements.

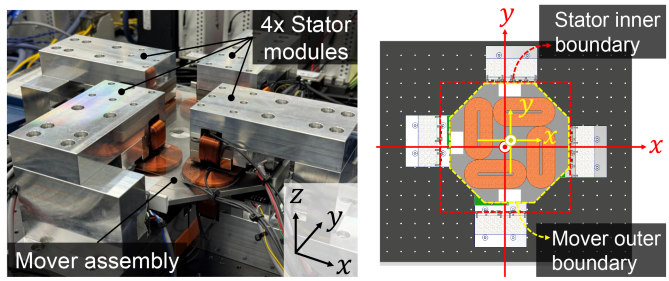
1) A high control frequency, together with a sufficiently long prediction horizon and deterministic timing, is required to improve tracking performance by reducing control delay and enabling long-horizon prediction of system behavior.
2) Both input and state constraints must be explicitly considered to keep the levitated mover within safe operating limits, including current, position, and velocity bounds.
3) Offset-free control is required to achieve high-precision servo performance, since model mismatches, fabrication tolerances, and power amplifier offsets can act as lumped disturbances and induce steady-state position errors.

One way to reduce computational load is to employ explicit MPC, which precomputes the optimal control law $u^* = \pi_{\mathrm{MPC}}(\boldsymbol{x})$ offline as a function of all feasible states $\boldsymbol{x}$. However, the main drawback of explicit MPC is that its computational complexity can grow exponentially with the number of states, constraints, and the prediction horizon [26]. Consequently, computing the explicit control law becomes infeasible for long-horizon MPC subject to state and input constraints, even in offline computation. Furthermore, even when the control law is obtained, identifying the corresponding region for a given system state can demand excessive processing power or memory, making real-time implementation difficult [27].

Another approach to reducing the computational burden of MPC is to approximate it using a deep neural network (DNN), introduced in [28], [29]. In this approach, the DNN is trained to learn the control law of MPC, enabling the control input to be computed directly without solving an optimization problem online, thereby significantly reducing the computational complexity of conventional MPC. In addition, DNNs are effective as they can naturally capture the piecewise-affine structure of explicit optimal control laws for constrained linear systems. This framework is further extended to FPGA-based implementations in [30], [31], [32], and the approximation error between the original MPC law and its DNN representation is analyzed and mitigated in [33].

However, despite the availability of this approach and the decouplable single-input single-output (SISO) structure of the system, prior studies have mainly focused on conceptual developments or applications to relatively simple hardware platforms. Consequently, the feasibility of applying this approach to physically actuated mechatronic and dynamic systems, including a 6-DOF magnetic levitation system, has not yet been experimentally investigated.

This article presents a DNN-approximated offset-free MPC framework for constrained in-plane motion control of a 6-DOF magnetic levitation system, with its FPGA implementation and experimental validation. By shifting the computational burden of MPC optimization from online to offline, the prediction horizon is no longer constrained by online computational resources. In addition, replacing the online MPC optimization with a DNN forward pass reduces the real-time computational burden, while FPGA implementation enables fast and deterministic execution. As a result, the proposed controller achieves 10 kHz control, which is ten times higher than that of previously reported MPC controllers for 6-DOF magnetic levitation systems [23], [24], [25], while providing offset-free tracking and explicit handling of input and state constraints.



(a) (b)

Fig. 1. (a) Photograph of the 6-DOF magnetic levitation system. (b) Cross-sectional view of the system in the xy-plane, showing the stator inner boundary and the mover outer boundary. The red and yellow coordinate frames denote the global and mover reference frames, respectively.

The contributions of this article are summarized as follows:

1) A 10-kHz offset-free MPC controller with reduced online computational cost is realized through DNN approximation and FPGA implementation, with explicit handling of hard input and softened state constraints.
2) An MPC framework tailored to the 6-DOF magnetic levitation system is proposed, integrating disturbance estimation, constrained control, DNN-based approximation, a stability analysis from a robust MPC perspective, and FPGA deployment.
3) Improved tracking accuracy and constraint satisfaction are demonstrated through experiments against linear controllers and prior MPC approaches.

The rest of the article is organized as follows. Section II provides an overview and dynamics model of the system. Section III presents the structure of the MPC framework. Section IV provides details of the DNN approximation and FPGA implementation. Section V presents experimental results validating the performance of the proposed control method. Finally, Section VI concludes the article.

II. System Dynamics Model

The 6-DOF magnetic levitation system used in this work is shown in Fig. 1(a), while Fig. 1(b) illustrates the coordinate definition and the mover positioned near the corner of the motion boundary $(+5\,\mathrm{mm}, +5\,\mathrm{mm})$. The system provides 6-DOF actuation using four Lorentz coil windings for in-plane motion along the x-, y-, and θ_z-DOFs, and four reluctance coil windings for out-of-plane levitation control of the z-, θ_x-, and θ_y-DOFs. This configuration results in a total of eight current inputs driven by eight power amplifiers (PAs).

From a control perspective, the system can be decoupled into six SISO linear subsystems with mutually independent control inputs and dynamics. Since planar motion is of primary importance in precision positioning applications, this work focuses on the x- and y-DOFs, which are controlled using the proposed MPC. The remaining four DOFs operate as magnetic bearings

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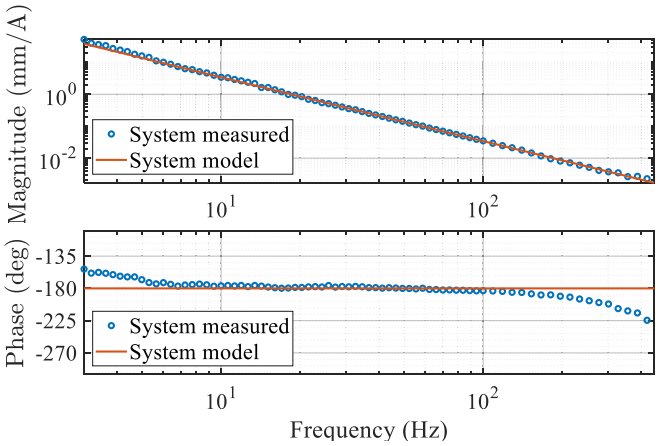


Fig. 2. Frequency response of the x-DOF of the system.

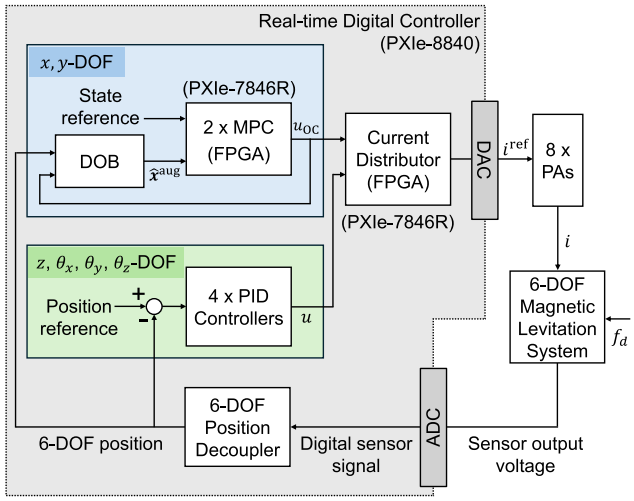


Fig. 3. Block diagram of the proposed MPC framework for the 6-DOF magnetic levitation system.

and are regulated by PID controllers [34] to maintain constant positions, making them less susceptible to constraint violations. Owing to the decoupled structure, the x- and y-DOF controllers can be designed and analyzed independently of these four DOFs. Furthermore, the geometric symmetry of the system makes the x- and y-DOFs dynamically identical, and therefore only the x-DOF is considered in the following analysis.

The x- and y-DOFs of the system can be modeled as a linear second-order free-mass system due to the noncontact and frictionless characteristics of the levitation mechanism. This is validated by the measured frequency response of the system, as shown in Fig. 2. Note that the dynamics of other 6-DOF magnetic levitation systems reported in [3], [4], [5], [8], [12], [24], [25] are also modeled in the same manner. Even when a system does not strictly exhibit free-mass dynamics, many force-driven physical systems can still be modeled as second-order systems. Therefore, the proposed method is applicable beyond magnetic levitation systems, demonstrating its generalizability across a wide range of motion control applications.

The discrete-time dynamics at time step k are given by

$$\boldsymbol{x}_{k+1} = A\boldsymbol{x}_k + Bu_k + Ef_d, \quad y_k = C\boldsymbol{x}_k \tag{1}$$

where $\boldsymbol{x}_k = [p_k,\, v_k]^\top \in \mathbb{R}^{2\times 1}$ denotes the state vector consisting of the mover's position and velocity, and u_k is the scalar control effort. The system matrix A, input matrix B, output matrix C, and disturbance matrix E are

$$A = \begin{bmatrix} 1 & t_s \\ 0 & 1 \end{bmatrix},\ B = \begin{bmatrix} \frac{K_{fx}}{2m}t_s^2 \\ \frac{K_{fx}}{m}t_s \end{bmatrix},\ E = \begin{bmatrix} \frac{1}{2m}t_s^2 \\ \frac{1}{m}t_s \end{bmatrix},\ C = \begin{bmatrix} 1 \\ 0 \end{bmatrix}^\top \tag{2}$$

where $t_s = 0.1$ ms, $K_{fx} = 34$ N/A, and $m = 2.52$ kg denote the sampling time, the force constant with respect to the control effort u_x, and the mover mass. The lumped disturbance f_d represents an equivalent disturbance force arising from fabrication tolerances, amplifier current offsets, model mismatch, and other unmodeled effects. In this form, the lumped disturbance f_d is assumed to be constant, as the mover's displacement varies only marginally within the prediction horizon at the kHz-level control rate [24]. Therefore, (1) can be augmented using the augmented state vector $\boldsymbol{x}_k^{\text{aug}} = \left[\boldsymbol{x}_k^\top, f_d\right]^\top \in \mathbb{R}^{3\times 1}$ as

$$\boldsymbol{x}_{k+1}^{\text{aug}} = A^{\text{aug}}\boldsymbol{x}_k^{\text{aug}} + B^{\text{aug}}u_k, \quad y_k = C^{\text{aug}}\boldsymbol{x}_k^{\text{aug}} \tag{3}$$

where the augmented system matrix A^{aug}, the input matrix B^{aug}, and the output matrix C^{aug} are given by

$$A^{\text{aug}} = \begin{bmatrix} 1 & t_s & \frac{1}{2m}t_s^2 \\ 0 & 1 & \frac{1}{m}t_s \\ 0 & 0 & 1 \end{bmatrix},\ B^{\text{aug}} = \begin{bmatrix} \frac{K_{fx}}{2m}t_s^2 \\ \frac{K_{fx}}{m}t_s \\ 0 \end{bmatrix},\ C^{\text{aug}} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}^\top. \tag{4}$$

III. MPC Structure

This section presents the MPC structure for the x- and y-DOF motion control. Fig. 3 shows the block diagram of the proposed 6-DOF magnetic levitation control system. The sensor signals acquired through the ADC are converted into 6-DOF positions by the position decoupler. The x- and y-DOFs are then controlled by the proposed DNN-approximated MPC controllers, implemented on the FPGA (PXIe-7846R), while the remaining four DOFs are regulated by PID controllers on the real-time controller (PXIe-8840). The resulting control efforts are mapped to eight coil current references and transmitted to the eight PAs through the DAC.

A. Input and State Constraints

The system constraints are divided into two categories: input and state constraints. First, the coil current limit of 7 A imposes constraints on the control input through the following relationship between the coil currents and the control inputs:

$$\begin{bmatrix} i_{L1} \\ i_{L2} \\ i_{L3} \\ i_{L4} \end{bmatrix} = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ -1 & 0 & 1 \\ 0 & -1 & 1 \end{bmatrix} \begin{bmatrix} u_x \\ u_y \\ u_{\theta_z} \end{bmatrix} \tag{5}$$

which is derived from the system topology and described in [34]. As this work primarily focuses on the x- and y-DOFs of the

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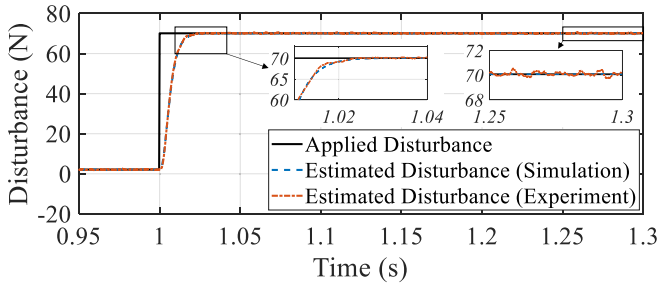


Fig. 4. Comparison of disturbance estimation between simulation and experiment under an additive 2 A input disturbance applied at $t = 1$ s.

system, the θ_z-DOF is regulated to a constant value and is used only for active stabilization, resulting in a small control input u_{θ_z} of approximately 1-2 A. Accordingly, the control input limits ($u_{\min}$ and $u_{\max}$) for the x- and y-DOFs are set to 5 A to provide a safety margin from the coil current limit.

The state constraints ($\boldsymbol{x}_{\min}$ and $\boldsymbol{x}_{\max}$) are divided into position and velocity constraints of the mover. Since the nominal travel range of the system is ± 5 mm in the xy-plane and the mover can travel up to ± 5.25 mm before the mover outer boundary contacts the stator inner boundary, the position constraint is set to ± 5.25 mm for both the x- and y-DOFs to prevent collisions with the stator structures. Velocity constraints are imposed for three purposes: satisfying application-specific speed limits, remaining within the measurable sensor range, and reducing vibration caused by excessive acceleration and deceleration. In this work, the velocity constraint is set to ± 200 mm/s.

B. Disturbance Observer (DOB) Design for State Estimation

A DOB is designed to estimate the augmented state $\boldsymbol{x}_k^{\text{aug}}$, as the sensing system provides only position measurements and not velocity or disturbance states. The structure of the DOB is formulated as

$$\hat{\boldsymbol{x}}_{k+1}^{\text{aug}} = A^{\text{aug}}\hat{\boldsymbol{x}}_k^{\text{aug}} + B^{\text{aug}}u_k + L^{\text{aug}}\left(y_k - C^{\text{aug}}\hat{\boldsymbol{x}}_k^{\text{aug}}\right) \tag{6}$$

where the hat symbol $\hat{\cdot}$ denotes an estimated value, and L^{aug} represents the observer gain matrix. The observer gain L^{aug} is designed by considering the tradeoff between estimation bandwidth and low-pass filtering characteristics. The resulting gain matrix is $L^{\text{aug}} = [0.137, 59.61, 21.15]^\top$, which places the continuous-time observer poles at -300, -500, and -600.

Fig. 4 compares the disturbance estimation performance between simulation and real-world experiments. The results show that the DOB accurately estimates the disturbance in both simulation and experiment. Note that even if mismatches exist in the force constant and mover mass, their effects are canceled during disturbance rejection within the offset-free control structure described in the following subsection.

C. MPC Formulation for Offset-Free Trajectory Tracking

The state prediction vector over the prediction horizon $N(=50)$ is defined as $X_k := [\boldsymbol{x}_{k+1}^\top, \boldsymbol{x}_{k+2}^\top, \ldots, \boldsymbol{x}_{k+N}^\top]^\top \in \mathbb{R}^{2N\times 1}$, and it can be predicted in terms of the input sequence vector $U_k := [u_k, u_{k+1}, \ldots, u_{k+N-1}]^\top \in \mathbb{R}^{N\times 1}$ as follows:

$$X_k = A_p\boldsymbol{x}_k + B_pU_k + E_pf_d \tag{7}$$

where the stacked prediction matrices are given by

$$A_p = \begin{bmatrix} A \\ A^2 \\ \vdots \\ A^N \end{bmatrix}, \quad B_p = \begin{bmatrix} B & 0 & \cdots & 0 \\ AB & B & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ A^{N-1}B & A^{N-2}B & \cdots & B \end{bmatrix}$$

$$E_p = \begin{bmatrix} E \\ AE + E \\ \vdots \\ A^{N-1}E + A^{N-2}E + \cdots + E \end{bmatrix}. \tag{8}$$

In real-time implementation, $\boldsymbol{x}_k$ and f_d are replaced by their DOB estimates $\hat{\boldsymbol{x}}_k$ and $\hat{f}_d$.

Next, the offset-free MPC cost function J_k is defined as

$$\begin{aligned} J_k := \sum_{i=0}^{N-1} \big[&(\boldsymbol{x}_{k+i} - \boldsymbol{x}_{k+i}^{\text{ref}})^\top Q(\boldsymbol{x}_{k+i} - \boldsymbol{x}_{k+i}^{\text{ref}}) \\ &+ (u_{k+i} - u_{k+i}^{\text{ss}})^\top R(u_{k+i} - u_{k+i}^{\text{ss}}) + \boldsymbol{s}_{k+i+1}^\top W\boldsymbol{s}_{k+i+1}\big] \\ &+ (\boldsymbol{x}_{k+N} - \boldsymbol{x}_{k+N}^{\text{ref}})^\top P(\boldsymbol{x}_{k+N} - \boldsymbol{x}_{k+N}^{\text{ref}}) \in \mathbb{R}^1 \end{aligned} \tag{9}$$

where $\boldsymbol{x}^{\text{ref}} \in \mathbb{R}^{2\times 1}$ and $\boldsymbol{s} \in \mathbb{R}^{2\times 1}$ denote the reference state and the nonnegative slack variable used to relax the state constraints, respectively. The steady-state control input u^{ss} is incorporated to ensure offset-free tracking performance and is analytically derived as $u^{\text{ss}} = -f_d/K_{fx}$ using (1) and (2). The weighting matrices for the state tracking error, control effort, terminal state, and slack variable are denoted by $Q \in \mathbb{R}^{2\times 2}$, $R \in \mathbb{R}^{1\times 1}$, $P \in \mathbb{R}^{2\times 2}$, and $W \in \mathbb{R}^{2\times 2}$, respectively, where $Q \succeq 0$, $R \succ 0$, and $W \succ 0$. In this work, these matrices are set as $Q = \text{diag}(10^4, 0.1)$, $R = 10$, and $W = 10^5\,\text{diag}(10^4, 0.1)$. The terminal weighting matrix P is selected as the solution of the discrete-time algebraic Riccati equation. These weights balance tracking performance and control effort. Without active constraints, the equivalent continuous-time closed-loop poles lie in the left-half complex plane, with a slightly overdamped response ($\zeta = 1.25$) and a closed-loop bandwidth of approximately 50 Hz, which is one-tenth of the mover's first structural resonance frequency. The slack weight penalizes constraint violations orders of magnitude more heavily than tracking errors while preserving feasibility. Note that these weights can be tuned differently depending on the control objective and the system dynamics.

The cost function in (9) can be compactly expressed as

$$\begin{aligned} J_k = &(\boldsymbol{x}_k - \boldsymbol{x}_k^{\text{ref}})^\top Q(\boldsymbol{x}_k - \boldsymbol{x}_k^{\text{ref}}) \\ &+ (X_k - X_k^{\text{ref}})^\top Q_N(X_k - X_k^{\text{ref}}) \\ &+ (U_k - U_k^{\text{ss}})^\top R_N(U_k - U_k^{\text{ss}}) + S_k^\top W_N S_k \end{aligned} \tag{10}$$

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where the stacked state reference, steady-state control effort, and slack variable vectors are defined as

$$X_k^{\text{ref}} := \begin{bmatrix} \boldsymbol{x}_{k+1}^{\text{ref}} \\ \boldsymbol{x}_{k+2}^{\text{ref}} \\ \vdots \\ \boldsymbol{x}_{k+N}^{\text{ref}} \end{bmatrix}, \; U_k^{\text{ss}} := \begin{bmatrix} u_k^{\text{ss}} \\ u_{k+1}^{\text{ss}} \\ \vdots \\ u_{k+N-1}^{\text{ss}} \end{bmatrix}, \; S_k := \begin{bmatrix} \boldsymbol{s}_{k+1} \\ \boldsymbol{s}_{k+2} \\ \vdots \\ \boldsymbol{s}_{k+N} \end{bmatrix} \tag{11}$$

and the block-diagonal weighting matrices are given by

$$\begin{aligned} Q_N &:= \text{diag}(Q, Q, \ldots, Q, P) \in \mathbb{R}^{2N\times 2N} \\ R_N &:= \text{diag}(R, R, \ldots, R, R) \in \mathbb{R}^{N\times N} \\ W_N &:= \text{diag}(W, W, \ldots, W, W) \in \mathbb{R}^{2N\times 2N}. \end{aligned} \tag{12}$$

To reduce the dimensionality of the learning inputs used in the subsequent learning process and to improve real-time computational efficiency, the proposed method does not use an explicit preview of the future reference sequence. Instead, a preview-free reference sequence is generated from the current reference state using an autonomous linear system, given by $\hat{X}_k^{\text{ref}} := A_p x_k^{\text{ref}}$. Then, (10) can be rewritten in the standard quadratic form with respect to the optimization vector $\boldsymbol{z}_k := \left[U_k^\top,\, S_k^\top\right]^\top \in \mathbb{R}^{3N\times 1}$ as

$$J_k = \frac{1}{2}\boldsymbol{z}_k^\top H_k \boldsymbol{z}_k + f_k^\top \boldsymbol{z}_k + C_k \tag{13}$$

where $H \in \mathbb{R}^{3N\times 3N}$ is the Hessian matrix and $f_k \in \mathbb{R}^{3N\times 1}$ is the first-order term, given by

$$\begin{aligned} H_k &= 2\,\text{diag}\left(B_p^\top Q_N B_p + R_N,\; W_N\right) \\ f_k &= \begin{bmatrix} 2B_p^\top Q_N \left(A_p(\boldsymbol{x}_k - \boldsymbol{x}_k^{\text{ref}}) + E_p f_d\right) - 2R_N U_k^{\text{ss}} \\ \mathbf{0}_{2N\times 1} \end{bmatrix}. \end{aligned} \tag{14}$$

The optimized vector $\boldsymbol{z}_k^*$ is obtained by solving the following quadratic programming (QP) problem with respect to the input and state constraints:

$$\begin{aligned} \boldsymbol{z}_k^* &= \arg\min_{\boldsymbol{z}_k} J(\boldsymbol{x}_k, f_d, \boldsymbol{x}_k^{\text{ref}}, \boldsymbol{z}_k) \\ \text{s.t.}\quad & u_{\min} \le u_{k+i} \le u_{\max} \\ & \boldsymbol{x}_{\min} - \boldsymbol{s}_{k+i+1} \le \boldsymbol{x}_{k+i+1} \le \boldsymbol{x}_{\max} + \boldsymbol{s}_{k+i+1} \\ & \boldsymbol{s}_{k+i+1} \ge \mathbf{0}_{2\times 1} \\ & \text{for } i = 0, 1, \ldots, N-1. \end{aligned} \tag{15}$$

Here, the input constraint is enforced as a hard constraint, whereas the position and velocity constraints are softened by the nonnegative slack variables to preserve the feasibility of the QP under state estimation noise.

Once the QP in (15) is solved, the optimized control input u_k^*, which is the first element of the optimized vector $\boldsymbol{z}_k^*$, is applied to the system, and the procedure is repeated in a receding horizon manner. Note that the mapping between the solution of (15) and the state is nonlinear and exhibits a piecewise-affine structure.

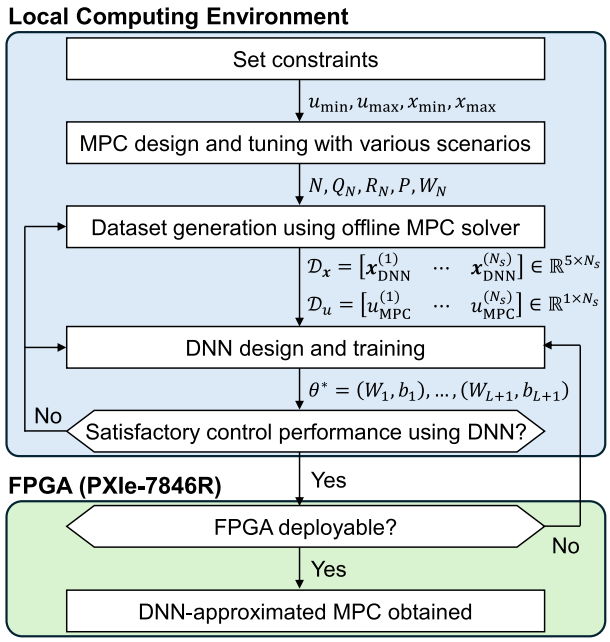


Fig. 5. Schematic flowchart of the design and implementation process of the proposed DNN-approximated MPC framework.

IV. DNN Approximation and FPGA Implementation

In this section, we describe the DNN structure used to approximate the MPC control law and its FPGA implementation. The overall controller design flowchart is shown in Fig. 5, consisting of an offline stage in the local computing environment and a deployment stage on the FPGA. In the offline stage, the constraints are specified and the MPC is designed and tuned. The MPC then serves as an offline solver to generate the state-input dataset $(\mathcal{D}_{\boldsymbol{x}}, \mathcal{D}_u)$ used to train the DNN parameters θ^*. The procedure returns to the MPC design step if the closed-loop performance with the trained DNN is unsatisfactory, or to the network design step if the FPGA resource or timing requirements are not met.

A. Dataset Generation

The input vector for the DNN is defined as

$$\boldsymbol{x}_{\text{DNN}} := \begin{bmatrix} p & v & f_d & p^{\text{ref}} & v^{\text{ref}} \end{bmatrix}^\top \in \mathbb{R}^{5\times 1} \tag{16}$$

where p, v, f_d, p^{ref}, and v^{ref} denote the mover position, mover velocity, lumped disturbance, position reference, and velocity reference, respectively. Note that during real-time control, p, v, and f_d are replaced by their estimated values using the DOB defined in (6). Next, to generate the training dataset, each component of the input vector $\boldsymbol{x}_{\text{DNN}}$ is sampled within finite ranges, with denser sampling near the constraint boundaries, as summarized in Table I. Then, a dataset $\mathcal{D}_{\boldsymbol{x}} = \left[\boldsymbol{x}_{\text{DNN}}^{(1)}, \boldsymbol{x}_{\text{DNN}}^{(2)}, \ldots, \boldsymbol{x}_{\text{DNN}}^{(N_s)}\right] \in \mathbb{R}^{5\times N_s}$ containing all sampled states is generated, where N_s denotes the total number of sampled state combinations obtained from the full Cartesian product of the sampling grids. The dataset $\mathcal{D}_{\boldsymbol{x}}$ covers a wide range of operating conditions expected to

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TABLE I
SAMPLING RANGES USED TO GENERATE THE DNN TRAINING INPUTS

Variable	Sampling Scheme
p (mm)	Dense near bounds: $[-5.5:0.1:-4.5,\ 4.5:0.1:5.5]$ Coarser in mid-range: $[-4.5:0.25:4.5]$
v (mm/s)	Dense near bounds: $[-220:4:-180,\ 180:4:220]$ Coarser in mid-range: $[-170:10:170]$
f_d (N)	Uniform sampling: $[-3:1:3]$
p^{ref} (mm)	Same sampling as p
v^{ref} (mm/s)	Same sampling as v with additional samples at $v^{\mathrm{ref}} = 0$

occur under various scenarios. Although this approach may include unlikely or unrealistic states and references, this conservative yet comprehensive sampling strategy covers a range beyond physically realizable operating conditions of the system. Therefore, it reduces the risk that the trained DNN produces control actions inconsistent with the original MPC policy, which could otherwise lead to unstable behavior in inherently unstable 6-DOF magnetic levitation systems. Next, the corresponding MPC control input for each sampled state is obtained by solving (15) offline on a local PC and is assembled into the dataset $\mathcal{D}_u = \left[u_{\mathrm{MPC}}^{(1)}, u_{\mathrm{MPC}}^{(2)}, \ldots, u_{\mathrm{MPC}}^{(N_s)}\right] \in \mathbb{R}^{1\times N_s}$. The resulting dataset $\mathcal{D}_u$ is used as training labels.

B. Offline MPC Approximation Via DNN

DNNs have recently become a popular choice for function approximation due to their universal approximation capability for highly nonlinear and high-dimensional mappings. From a control perspective, DNNs can effectively capture the complex nonlinear relationships inherent in the MPC control law while incurring low computational load once trained.

In this work, the network takes the input vector $\boldsymbol{x}_{\mathrm{DNN}}$ and produces a scalar control output $u_{\mathrm{DNN}} = \pi_{\mathrm{DNN}}(\boldsymbol{x}_{\mathrm{DNN}}; \theta)$. For a DNN with L hidden layers, the mapping π_{DNN} is expressed as the composition

$$\pi_{\mathrm{DNN}}(\boldsymbol{x}_{\mathrm{DNN}}; \theta) = f_{L+1} \circ f_L \circ \cdots \circ f_1(\boldsymbol{x}_{\mathrm{DNN}}) \tag{17}$$

where each hidden layer performs an affine transformation followed by a nonlinear activation,

$$f_l(z_{l-1}) = \sigma(W_l z_{l-1} + b_l), \quad l = 1, \ldots, L \tag{18}$$

and the output layer is given by

$$f_{L+1}(z_L) = W_{L+1} z_L + b_{L+1}. \tag{19}$$

Here, $z_0 = \boldsymbol{x}_{\mathrm{DNN}} \in \mathbb{R}^{5\times 1}$, $z_l \in \mathbb{R}^{H_l \times 1}$ for $l = 1, \ldots, L$, $W_l \in \mathbb{R}^{H_l \times H_{l-1}}$, and $b_l \in \mathbb{R}^{H_l \times 1}$. All trainable parameters are collected as $\theta := \{W_l, b_l\}_{l=1}^{L+1}$.

For the activation function, the rectified linear unit (ReLU), defined as $\sigma(z) = \max(0, z)$, is employed in all hidden layers and applied elementwise. With ReLU activations, the resulting network mapping is piecewise-affine, which is compatible with the piecewise-affine character of explicit MPC laws. Moreover, a ReLU-based DNN is well suited for FPGA deployment due to the low computational complexity and simple hardware implementation of ReLU operations.

TABLE II
ARCHITECTURE OF THE EMPLOYED DNN

Layer	Type	Neurons	Activation
Input layer	Input	5	–
Hidden layer 1-2	Fully connected	32	ReLU
Hidden layer 3-4	Fully connected	8	ReLU
Output layer	Fully connected	1	Linear

TABLE III
TRAINING HYPERPARAMETERS OF THE DNN

Hyperparameter	Value
Optimizer	Adam
Initial learning rate	5×10^{-3}
Learning rate schedule	Piecewise decay
Learning rate drop period	30 epochs
Learning rate drop factor	0.95
Mini-batch size	524,288

Detailed information on the employed DNN is presented in Table II. A gradually narrowing structure is adopted, where wider layers improve approximation capability and narrower layers reduce the number of parameters for efficient FPGA implementation. The network is designed to be as large as permitted by the available FPGA resources. Note that the proposed framework is not restricted to this architecture, and other network sizes and structures can be employed as long as sufficient approximation accuracy is achieved.

The DNN is trained in a supervised manner using data generated from the MPC controller. Let N_s denote the number of training samples, and let $\{(\boldsymbol{x}_{\mathrm{DNN}}^{(i)}, u_{\mathrm{MPC}}^{(i)})\}_{i=1}^{N_s}$ be the input–output pairs, where $u_{\mathrm{MPC}}^{(i)}$ is the control input computed by the MPC for the ith sample. The training problem is then formulated using the mean squared error (MSE) loss as

$$\theta^* = \arg\min_{\theta} \frac{1}{N_s} \sum_{i=1}^{N_s} \left(\pi_{\mathrm{DNN}}(\boldsymbol{x}_{\mathrm{DNN}}^{(i)}; \theta) - u_{\mathrm{MPC}}^{(i)}\right)^2. \tag{20}$$

This supervised learning formulation enables the DNN to approximate the MPC control law over the sampled state space. The training hyperparameters are summarized in Table III.

The approximation result of the trained DNN is evaluated over the generated dataset. Fig. 6 presents a representative control surface near the position limit, demonstrating that the trained DNN closely reproduces the nonlinear characteristics of the MPC control law. Quantitatively, the root mean square error (RMSE), mean error, and normalized RMSE (NRMSE) are 0.0756 A, -2.49×10^{-5} A, and 0.0151, respectively. Here, NRMSE is defined as the RMSE normalized by the maximum absolute value of the MPC control input. The obtained errors are below the level of the inherent current noise, indicating that the approximation accuracy is sufficient for practical implementation.

C. DNN Steady-State Offset Compensation

The approximation error of the trained network is small but generally nonzero due to the finite size of the training dataset

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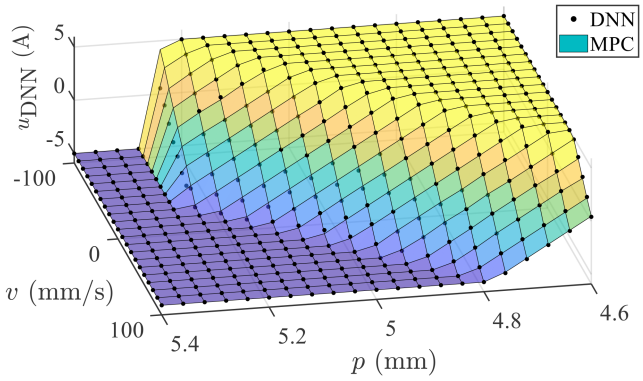


Fig. 6. Approximation of the MPC control surface by the trained DNN over the (p, v) state space for $f_d = 0.5$ N, $p^{\mathrm{ref}} = 5$ mm, and $v^{\mathrm{ref}} = 0$ mm/s, demonstrating accurate approximation.

and the network. To mitigate the resulting steady-state error, an offset compensation method, inspired by [33], is employed and adapted to the system.

In this approach, the presaturation offset-compensated control input $u_{\mathrm{OC}}^{\mathrm{pre}}$ is defined as

$$\begin{aligned} u_{\mathrm{OC}}^{\mathrm{pre}} &:= u_{\mathrm{DNN}} + (u^{\mathrm{ss}} - u_{\mathrm{DNN}}^{\mathrm{ss}}) \\ &= \pi_{\mathrm{DNN}}(\boldsymbol{x}_{\mathrm{DNN}}) + \left(-\hat{f}_d/K_{fx} - \pi_{\mathrm{DNN}}(\boldsymbol{x}_{\mathrm{DNN}}^{\mathrm{ss}})\right) \end{aligned} \tag{21}$$

and the final control input is additionally clipped to enforce the input constraint more conservatively as

$$u_{\mathrm{OC}} = \min\left(u_{\max}, \max\left(u_{\min}, u_{\mathrm{OC}}^{\mathrm{pre}}\right)\right) \tag{22}$$

where $\boldsymbol{x}_{\mathrm{DNN}}^{ss}$ denotes the steady-state DNN input vector, defined as $\boldsymbol{x}_{\mathrm{DNN}}^{ss} := \left[p^{\mathrm{ref}},\ 0,\ \hat{f}_d,\ p^{\mathrm{ref}},\ 0\right]^{\top}$, and u_{DNN}^{ss} is the output of the same trained DNN evaluated at $\boldsymbol{x}_{\mathrm{DNN}}^{ss}$.

In this formulation, the same DNN evaluated at the steady-state reference, corresponding to zero reference velocity, is used to compensate for the steady-state error of the trained DNN. This auxiliary steady-state evaluation avoids the need for a separately trained secondary network and effectively reduces steady-state errors without increasing network size.

D. Closed-Loop Stability of DNN-Approximated MPC

The deployed control input u_{OC} deviates from the nominal MPC law π_{MPC} owing to the DNN approximation, the steady-state offset compensation, and the input clipping. We treat this deviation, $d := u_{\mathrm{OC}} - \pi_{\mathrm{MPC}}(\boldsymbol{x}_{\mathrm{DNN}})$, as a bounded input perturbation, i.e., $u_{\mathrm{OC}} = \pi_{\mathrm{MPC}}(\boldsymbol{x}_{\mathrm{DNN}}) + d$, and provide a statistical validation following [29].

Assumption 1: The closed-loop system under input $u_{\mathrm{OC}} = \pi_{\mathrm{MPC}}(\boldsymbol{x}_{\mathrm{DNN}}) + d$ is stable for all initial conditions $\boldsymbol{x}_{\mathrm{DNN}}(0)$ over the feasible region $\mathcal{X}_{\mathrm{feas}} \subset \mathbb{R}^5$ under perturbations satisfying $|d| \le \eta$. This is verified empirically by closed-loop simulation under worst-case perturbations $|d| = \eta$.

Lemma 1: Let $\tilde{\mu}$ be the empirical satisfaction rate of $|u_{\mathrm{OC}} - \pi_{\mathrm{MPC}}(\boldsymbol{x}_{\mathrm{DNN}})| \le \eta$ along the closed-loop trajectory $\mathcal{X}_i$ from $\boldsymbol{x}_{\mathrm{DNN}}(0)$ over a finite horizon, evaluated over M i.i.d. initial conditions $\boldsymbol{x}_{\mathrm{DNN}}(0) \sim \mathcal{X}_{\mathrm{feas}}$. If $\tilde{\mu} - \varepsilon_h \ge \mu_{\mathrm{crit}}$ with $\varepsilon_h = \sqrt{\ln(2/\delta_h)/(2M)}$, then with confidence at least $1 - \delta_h$,

$$\begin{aligned} \mathbb{P}_{\boldsymbol{x}_{\mathrm{DNN}}(0)\sim\mathcal{X}_{\mathrm{feas}}}\big(\,&|u_{\mathrm{OC}} - \pi_{\mathrm{MPC}}(\boldsymbol{x}_{\mathrm{DNN}})| \le \eta, \\ &\forall \boldsymbol{x}_{\mathrm{DNN}} \in \mathcal{X}_i\big) \ge \mu_{\mathrm{crit}}. \end{aligned} \tag{23}$$

With Assumption 1, this provides statistical evidence of closed-loop stability with probability at least μ_{crit} over $\mathcal{X}_{\mathrm{feas}}$.

Proof: This follows directly from Hoeffding's inequality, applied as in [29]. ■

Remark 1: We set $\eta = 0.5$ A, $\delta_h = 0.01$, and $\mu_{\mathrm{crit}} = 0.99$. Validation with $M = 200\,000$ trajectories yields $\tilde{\mu} = 0.9949$ and $\tilde{\mu} - \varepsilon_h = 0.9912 \ge \mu_{\mathrm{crit}}$, confirming closed-loop stability with probability at least 99% at 99% confidence. As Assumption 1 is empirical, this is a statistical validation rather than a constructive robust-stability certificate.

E. FPGA Implementation of the DNN for 10 khz Control

Achieving fixed computation latency is challenging in CPU-based implementations due to timing jitter from operating-system scheduling, interrupts, and memory effects, whereas FPGA enables cycle-level deterministic execution. Therefore, the trained networks are implemented on a single FPGA module for deterministic 10-kHz real-time operation. To eliminate external-memory latency and maintain consistent execution time, all network parameters are stored in on-chip memory.

Four identical networks are implemented in parallel for offset-compensated xy-DOF control, each mapped onto the FPGA as a sequence of matrix–vector multiply–accumulate (MAC) operations followed by elementwise ReLU activations. A pipelined MAC datapath is time-multiplexed across the neurons and layers of that network, with each DSP block configured to perform one MAC per clock cycle.

For efficient hardware implementation, the network is quantized to a 24-bit signed fixed-point format (1 sign bit, 8 integer bits, and 15 fractional bits). All weights, biases, and intermediate activations are represented and processed using this format throughout the forward computation. The representable range is empirically verified to cover all trained weights and biases and all intermediate activations observed when evaluating the network over the fully generated dataset. Saturation arithmetic is adopted to avoid wrap-around effects in the control signal. The resulting quantization-induced error over the full dataset remains small (RMSE: 0.0045 A), indicating that the fixed-point representation preserves the floating-point network output with negligible degradation.

Consequently, at each control cycle, the DNN input vector $\boldsymbol{x}_{\mathrm{DNN}}$ is updated via a register-based interface, and the offset-compensated control input u_{OC} is computed on the FPGA and converted to the analog current reference through a DAC.

V. Experimental Validation

A. System Integration and Hardware Setup

The overall system is implemented on a National Instruments PXIe platform, which is widely used for high-precision mechatronic control systems [2], [6], [13], [23], [34]. The trained

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TABLE IV
FPGA RESOURCE UTILIZATION OF THE PXIE-7846R FOR THE PROPOSED 6-DOF CONTROL SYSTEM

Resource	Used	Available	Utilization (%)
Slices	23,192	25,350	91.5
Registers	92,105	202,800	45.4
LUTs	59,362	101,400	58.5
Block RAMs	20	325	6.2
DSP48 Blocks	138	600	23.0

DNNs are synthesized and deployed on an FPGA module (PXIe-7846R, National Instruments), operating at 40 MHz. The FPGA also acquires position signals from the optical reflective sensors, which are then decoupled into six DOFs based on the system geometry. Hardware resource utilization for the implemented FPGA is summarized in Table IV. The synthesized design meets timing closure at 40 MHz with 91.5% slice utilization, meaning that all critical paths satisfy the 25 ns clock period with no setup or hold violations.

Real-time operations outside the FPGA–including control algorithms for the remaining four DOFs (z, θ_x, θ_y, and θ_z), as well as the LabVIEW-based supervisory interface and data logging modules–are implemented on a CPU-based real-time controller (PXIe-8840, National Instruments) and executed at 10 kHz. Note that the proposed control framework can be implemented on other FPGA platforms and control environments.

B. Controllers for Comparative Experiments

Six motion controllers (C0–C5) are compared in the following experiments, and their details are summarized below. Among them, C1-C5 are implemented on the CPU-based real-time controller (PXIe-8840).

C0. The proposed MPC (10 kHz): The control input defined in (22) is computed on the FPGA and applied to the system.

C1. PID controller (10 kHz): The controller is implemented in a lead–lag compensator form, defined as

$$u_c(s) = \mathrm{sat}(C(s)e(s)),\ C(s) = K_p \frac{\alpha\tau s + 1}{\tau s + 1}\frac{T_i s + 1}{T_i s}. \quad (24)$$

Here, the saturation function is defined as $\mathrm{sat}(u) = \min(u_{\max}, \max(u_{\min}, u))$, and $e(s)$ denotes the position error. The controller parameters K_p, α, τ, and T_i are set to 6.1 A/mm, 10, 0.503 ms, and 53.05 ms, respectively, which are tuned to achieve a crossover frequency of 70 Hz.

C2. LQR controller (10 kHz): For a fair comparison, the weighting matrices Q and R are chosen to be identical to those of C0, and the same saturation function as in C1 is applied. The observer used in C0 is also employed for state estimation.

C3. Baseline MPC controller (1 kHz): The controller employs a standard linear MPC formulation, optimizing control inputs over a prediction horizon, subject to input constraints.

C4. Offset-free MPC in [24] (1 kHz): The prediction model is augmented with the lumped disturbance estimated by the DOB to eliminate steady-state errors.

C5. Predictor-based MPC in [25] (1 kHz): An FSA-based predictor is incorporated to compensate for system delays, enabling the controller to account for future states during optimization. For a fair comparison, C3-C5 use the same weighting matrices (Q and R) as C0. The observer poles are matched to those of C0, ensuring identical observer dynamics.

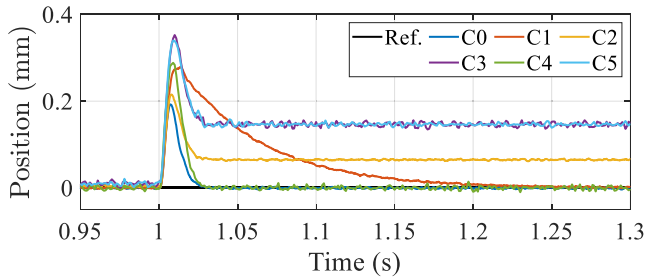


Fig. 7. Experimentally measured disturbance rejection responses of C0-C5 to an additive 2 A input disturbance applied at $t = 1$ s.

The computational times for C3-C5 are measured to be 711 μs on average, with a standard deviation of 30.9 μs, indicating that 1 kHz is the maximum practical control frequency under the available computational resources. When state constraints are additionally imposed, the online MPC is unable to consistently satisfy the 1 ms deadline. In contrast, the DNN forward-pass latency of C0 for full x- and y-DOF control is measured to be 37 μs, which is approximately 5% of online MPC's computation time. It also operates deterministically with zero standard deviation, enabling reliable 10-kHz real-time execution.

C. Comparative Experiments

To evaluate the motion performance and constraint-handling behavior of the proposed MPC framework, three comparative experiments are conducted using controllers C0–C5, including disturbance rejection, step response, and trajectory tracking.

1) Disturbance Rejection: To evaluate the disturbance rejection capability achieved by integrating the DOB into the prediction model, a 2 A step disturbance is injected additively into the control input at $t = 1$ s. The experimental results in Fig. 7 show that the controllers incorporating the DOB, C0 and C4, can effectively eliminate steady-state errors, confirming the DOB's disturbance rejection capability. The integral controller in C1 can remove steady-state error, but its passive response to accumulated error results in slower convergence. Increasing the integral gain accelerates error elimination but degrades phase margin and stability while increasing overshoot, whereas DOB-incorporated controllers reduce this tradeoff.

2) Step Response: To evaluate the transient performance of the proposed MPC framework, step response tests are conducted with step sizes of 0.5 and 5.0 mm. The small-step test examines the transient response under active input constraints only, whereas the large-step test evaluates the transient response when both input and state constraints are active. A step reference is applied to the x-DOF at $t = 1$ s. The experimental results for 0.5 and 5.0 mm steps are shown in Fig. 8 and Fig. 9, respectively, and the corresponding performance metrics are summarized in Table V.

During the small-step test, the 5% settling time, steady-state error, and steady-state position variation are evaluated as performance indices. The steady-state error is defined as the average tracking error over 2.5–3 s, while the steady-state position

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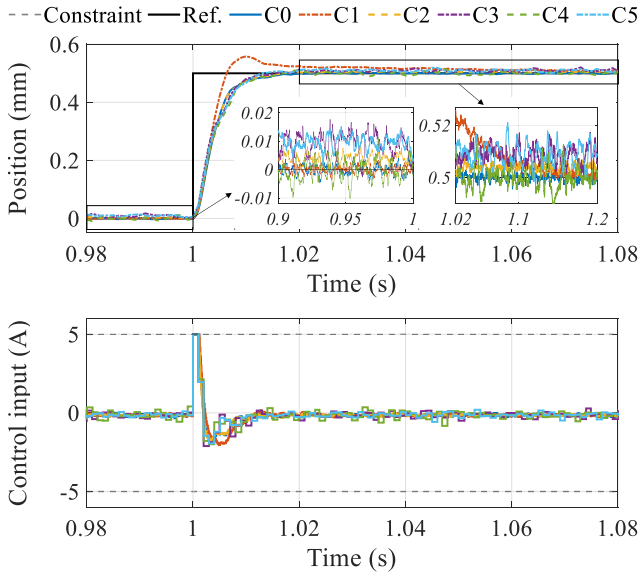


Fig. 8. Experimental results of 0.5-mm small-step responses and control inputs for C0-C5 under input constraints.

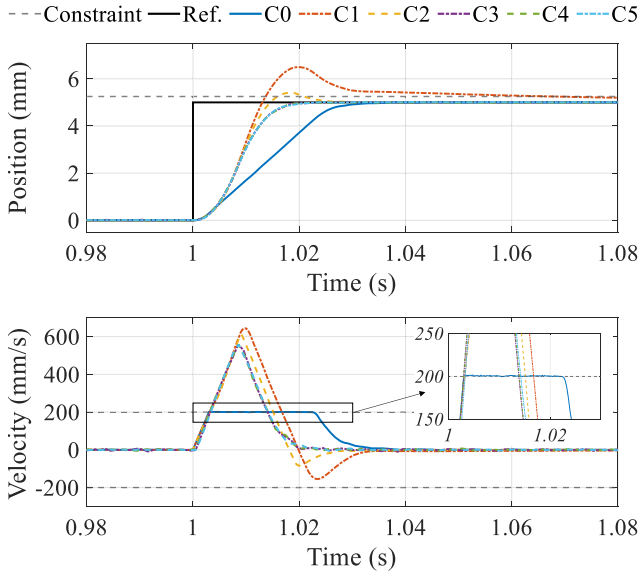


Fig. 9. Experimental results of 5.0-mm large-step response comparison of C0-C5 under input, position, and velocity constraints.

variation is defined as the standard deviation over the same interval. As summarized in Table V, C0 provides the best overall performance in terms of transient response and steady-state accuracy, owing to its DOB-incorporated structure and optimal control capability. In contrast, the steady-state error observed in C2, C3, and C5 can be attributed to a lumped disturbance caused by fabrication tolerances and power amplifier offsets. Moreover, the steady-state position variation of the 1 kHz controllers (C3-C5) is significantly larger than that of the 10 kHz controllers, as the higher sampling rate reduces ZOH-induced delay and phase lag.

During the large-step test, the 5.0 mm step reference activates both input and state constraints, and the maximum position and velocity are evaluated accordingly. For hardware protection, the measured position signal is artificially shifted by 2.5 mm during near-boundary tests to prevent a physical collision with the stator structure. Consequently, when the physical mover position is 2.75 mm, the controller treats it as 5.25 mm, creating a virtual safety margin.

Under this condition, C0 maintains safe operation near the position limit and keeps the velocity close to its limit, even under the large-step reference. The small velocity excursions ($\leq$ 1.5 mm/s) are attributed to sensor noise and the soft-constraint policy. To quantify the two contributions, the same scenario is reproduced in a noise-free simulation with the MPC law of (15), which yields a velocity slack of only $s^*_{v,\max} = 0.064$ mm/s. The excursion is thus dominated by the DOB velocity estimation noise, whose standard deviation, measured over a 1 s interval, is $\sigma_v = 0.482$ mm/s, giving a 3σ bound of 1.45 mm/s that accounts for the remaining 1.44 mm/s. In contrast, classical controllers C1 and C2 exceed the allowable range of motion due to the lack of explicit constraint handling. Although C3–C5 satisfy the position limit, they enforce only input constraints due to the real-time computational limitations of online MPC and therefore substantially exceed the velocity constraint. Consequently, only the proposed controller simultaneously respects the hard input limit and the softened state constraints within the quantified tolerance, whereas the other controllers violate the state limits by a large margin.

To confirm that the artificial shift does not affect the validity of the constraint-handling results, an additional large-step test is conducted with C0 without the 2.5 mm shift. The mover reaches a maximum position of 5.01 mm and a maximum velocity of 201.6 mm/s, showing agreement with the shifted case, without any contact with the stator structure.

3) Trajectory Tracking: To evaluate the trajectory tracking performance of the proposed MPC framework, an x-DOF slope tracking test and an xy-plane zigzag trajectory tracking test are conducted, and the corresponding experimental results are presented in Fig. 10, Fig. 11, and Table VI.

TABLE V
STEP RESPONSE TEST RESULTS

Step size	Metric	C0	C1	C2	C3	C4	C5
0.5 mm	T_s (ms)	10.1	17.2	9.9	9.7	11.2	10.5
	e_{ss} (μm)	0.01	0.00	-4.13	-9.27	-0.02	-9.17
	σ_{ss} (μm)	0.92	0.84	0.96	5.18	5.56	4.73
5.0 mm	p_{max} (mm)	5.01	6.50	5.41	5.04	5.03	5.02
	v_{max} (mm/s)	201.5	644.2	610.4	557.8	558.1	559.0
	e_{ss} (μm)	0.00	-0.01	-4.13	-8.70	-0.03	-8.93

TABLE VI
TRAJECTORY TRACKING TEST RESULTS

Case	Metric	C0	C1	C2	C3	C4	C5
Slope tracking (170 mm/s)	e_{rms} (μm)	69.0	144.7	95.2	150.7	141.2	133.8
	e_{max} (μm)	236.5	275.5	240.8	441.6	422.7	422.3
	p_{max} (mm)	5.01	5.38	5.27	5.25	5.24	5.16
	v_{max} (mm/s)	200.6	232.2	220.5	227.2	231.8	231.2
Zigzag tracking	e_{rms} (μm)	34.6	137.8	38.5	130.5	133.4	118.6
	e_{max} (μm)	141.7	352.9	141.5	370.7	371.5	352.5
	p_{max} (mm)	5.02	5.32	5.09	5.11	5.10	5.11
	v_{max} (mm/s)	200.3	248.8	202.8	226.9	211.0	218.7

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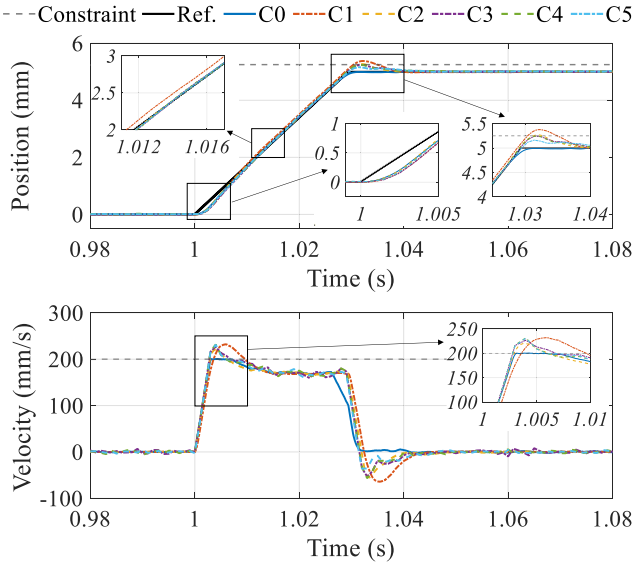


Fig. 10. Experimentally measured x-DOF slope trajectory tracking results, where the position and velocity references are within their limits.

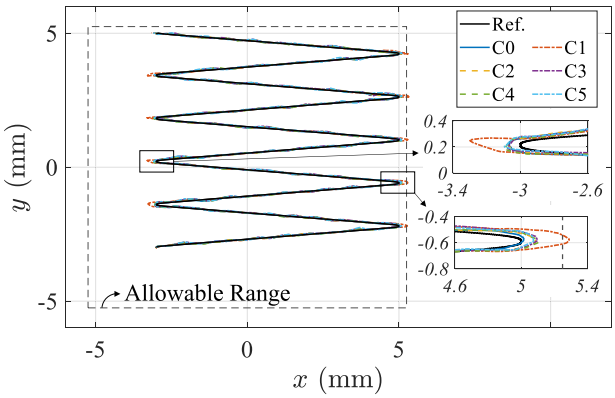


Fig. 11. Experimentally measured in-plane zigzag trajectory tracking results in the xy-plane, with the allowable motion range.

In the slope tracking tests, the mover is commanded to move from 0.0 mm to +5.0 mm with a velocity of 170 mm/s, both of which are within the specified position and velocity constraints. The tracking root-mean-square error e_{rms} over the interval $t = 1.00$-1.05 s, maximum absolute error e_{max}, maximum position p_{max}, and maximum velocity v_{max} are evaluated as performance metrics. As in the previous tests, the measured position is artificially shifted by 2.5 mm to avoid a collision with the stator structure. Under this condition, only C0 satisfies both the position and velocity constraints within the allowable range, whereas the other controllers violate at least one of these limits. This result highlights the effectiveness of C0 in achieving constraint-aware high-speed tracking without compromising safety. The same slope-tracking test is additionally performed with C0 without the 2.5 mm shift, resulting in $e_{\text{rms}} = 78.0\,\mu$m, $e_{\text{max}} = 240.2\,\mu$m, $p_{\text{max}} = 5.03$ mm, and $v_{\text{max}} = 200.9$ mm/s, which confirms that C0 keeps the position within the ± 5.25 mm limit and the velocity close to the 200 mm/s limit even without the virtual safety margin.

Next, a tracking test using a conventional zigzag trajectory, which is widely used in scanning applications [35], is conducted to demonstrate the 2-DOF tracking capability in the xy-plane. The reference trajectory in the x-DOF is generated using a fourth-order polynomial with an 8 mm travel distance, a maximum velocity of 170 mm/s, and a maximum acceleration of 5 G, all within the specified input, position, and velocity constraints. For the y-DOF, a constant-velocity reference is used over a travel distance of 8 mm. The measured positions in both the x- and y-DOFs are artificially shifted by 1 mm. For performance evaluation, the root-mean-square tracking error e_{rms} and maximum absolute error e_{max} in the xy-plane, as well as the maximum position p_{max} and maximum velocity v_{max} in the x-DOF, are used. Fig. 11 presents the corresponding tracking results, indicating that C0 behaves more conservatively as x approaches 5.0 mm due to its explicit consideration of the position constraints. As summarized in Table VI, the proposed controller C0 achieves the best overall trajectory-tracking accuracy while effectively respecting the imposed position and velocity limits. Likewise, the zigzag trajectory test repeated with C0 without the 1 mm shift results in $e_{\text{rms}} = 35.3\,\mu$m, $e_{\text{max}} = 126.4\,\mu$m, $p_{\text{max}} = 5.02$ mm, and $v_{\text{max}} = 199.3$ mm/s, verifying safe operation at the true physical boundary.

VI. Conclusion

In this article, a DNN-approximated offset-free MPC framework for constrained in-plane control of a 6-DOF magnetic levitation system has been presented and implemented on an FPGA to achieve three primary research objectives: 1) enabling 10-kHz real-time control with a sufficiently long prediction horizon and deterministic timing, 2) explicitly handling hard input and softened state constraints, and 3) achieving offset-free tracking. To accomplish these goals, the MPC control law is approximated by a DNN and deployed on an FPGA, shifting optimization offline and enabling fast and deterministic computation. The resulting controller operates at a control frequency of 10 kHz, which is ten times higher than those reported in prior studies on MPC-based control of 6-DOF magnetic levitation systems. To validate the proposed MPC framework, comparative experiments are conducted using various motion controllers. The results show that the proposed MPC approach achieves the best overall tradeoff among disturbance rejection, constraint handling, and tracking accuracy. These results demonstrate the potential of the proposed MPC framework for industrial and academic applications requiring high-precision 6-DOF magnetic levitation motion control.

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